

Submission Summary

Conference Name

2026 5th International Conference on Data, Decision and Systems

Paper ID

108

Paper Title

A Hybrid Quantum-Classical Framework for Spatiotemporal Flood Risk Assessment and Emergency Response Planning

Abstract

Coastal Dakshina Kannada, Karnataka, experiences recurring monsoon flooding driven by river discharge and tidal backwater interaction. Sentinel-1 SAR revisits the region only once every twelve days, limiting direct flood-event mapping. This paper presents a hybrid quantum-classical framework to address this constraint. A static flood-susceptibility model trained on a 20-pass SAR flood-frequency inventory and nine terrain and hydrological factors achieved a test ROC-AUC of 0.9625 using XGBoost. A rainfall- and tide-based temporal trigger dynamically modulates the risk surface. The fused map enables flood-aware routing on Mangaluru's road network, where A* explored 31.3% fewer nodes than Dijkstra while returning identical optimal paths. Under May 2025 conditions, flood-aware routing reduced mean exposure by 54% with a 3.8-minute detour. A 9-qubit QUBO ambulance-dispatch model solved using QAOA achieved a 0.925–0.926 approximation ratio and was verified against the Hungarian algorithm.

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Primary Subject Area

Applications

Submission Files

conference_paper.pdf (218.6 Kb, 2026-08-05 16:45:24 GMT+5:30)
